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FarmNodes & Distributed Research Farms

Governed virtual laboratories, distributed execution and independent verification

BTG-LIB-FARMNODES-SG · Edition 1.0 · Audience: Research sponsors, laboratory operators, institutions, communities and reviewers · Issued 23 October 2026

Publication boundary: This is an explanation of the system and its public operating model, not a source release, engineering specification, certification or promise of live service availability. Product status is stated with the limitations below.

Executive overview

A FarmNode is a governed computational research laboratory defined by its identity, problem domain, permissions, tools, Circuits, computing limits and evidence rules. A farm may operate on one or more physical hosts, but logical identities are not proof of independent computers or institutions.

01 — The laboratory concept

A traditional laboratory contains a purpose, instruments, methods, trained people, records and limits. A BTG FarmNode models many of those roles computationally for suitable research. It can preserve which problems were assigned, what resources were used and what tests evaluated the result.

Farms may focus on robotics, agriculture, infrastructure, software, mathematics, environmental research or other authorized domains. A domain name does not imply that every specialized tool or dataset is present.

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02 — Identity and authority

Each research farm should have a distinct identified mandate, responsible owner or operator, accepted capabilities and specific access to input information. An organization sponsoring an experiment is not automatically authorized to read every record held by another farm.

ENTITY can record authority and provenance. ADAM governs permitted protected state changes. A signed transport receipt shows delivery, not scientific agreement, legal ownership or independent reproduction.

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03 — Research task lifecycle

A research sponsor defines the question, success criterion, available permitted inputs, resource budget, time limits, privacy limits and responsible reviewers. A farm selects qualified tools and may use NIKI to propose candidate approaches.

Outputs should be accompanied by failed tests, version references, measurement conditions and an evaluation report. Only after independent review and rights assessment could a result be considered for reuse or commercial treatment.

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04 — Virtual versus physical independence

Thousands of registered laboratory definitions or logical bindings may exist on a single machine. Multiple isolated processes on that machine demonstrate local orchestration, but not independence between physical hosts or organizations.

True independent multi-node qualification requires distinct host and operator controls, separate keys and storage as appropriate, failures of one host not compromising all others, and tests showing reproducibility across environments.

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05 — Collaboration between farms

A permitted evidence capsule can describe a problem, input lineage, test requirements and non-sensitive result without forwarding every raw data source. Another laboratory may independently challenge a claim or produce an alternative solution.

Reviewers should distinguish delivery, rerun of identical binaries, independently authored reproduction and independent scientific validation. Each has a different level of evidentiary strength.

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06 — Resource limits and responsible use

Farm work can consume CPU, memory, storage, bandwidth and specialized hardware. The operating mandate should include budgets and shutoff conditions, alongside safety restrictions and applicable policies.

An automated research workflow must not be allowed to run unauthorized software, reach protected networks or disclose sensitive material simply because it can propose a useful experiment.

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07 — Evidence and failed experiments

A credible farm retains original hypotheses, test oracles where releasable, tool versions, errors, negative results and changes made after failed attempts. This prevents cherry-picked demonstrations from being presented as universal capability.

Audit reviewers should be able to verify what occurred on a specific host in a specified period. Metrics taken from a local development workstation cannot be described as outcomes of independent research institutions.

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08 — Potential regional and institutional value

A distributed research arrangement could allow smaller communities to contribute qualified technical work and collaborate with universities or manufacturers without physically relocating all expertise.

Actual economic benefit depends on trained participants, customer demand, rights clearance, sustainable operating costs, access to equipment and external acceptance. Research potential is not realized employment or booked income.

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09 — From finding to economic asset

A successful FarmNode output might become a validated dataset, algorithm, Circuit, industrial method or report. BTDU can retain governed representations and ENTITY can preserve relevant rights claims and evidence.

The proposed Exchange may later evaluate a rights-defined commercial or investment instrument. A lab result is not automatically a listed security, and contributors need lawful arrangements to receive any economic entitlement.

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10 — Current programme qualification

Public engineering records distinguish local functioning research cycles from qualified deployments on genuinely separate physical machines. Three independent FarmNode hosts were an outstanding external demonstration requirement in the latest published records.

Any future production claim must identify the exact hosts, operating period, identity controls, failure tests and third-party reviewers, rather than counting logical registry objects as machines.

Related manuals and use

The companion User & Operator Manual describes the decisions and reviews applicable to this system. References to proposed user flows are explicitly conditional on qualification of the relevant product release.